Midterm 1 - Fall 2024

ECON 4753 — University of Arkansas

1. This question concerns our general discussion of modelling outcomes. Say we have an outcome variable Y that we want to analyze and a set of variables X we want to use in our model. In general, we write the 'true model' as $Y_i = f(X_i) + e_i$. With a sample, we will fit some model which we will call $\hat{f}$.

   Two models were fit using the same training sample. Model A is a simple linear regression, while Model B allows a more complicated relationship between X and Y.

Model	In-sample MSPE
A	420
B	310

   i. Write the formula for in-sample mean-squared prediction error.

   ii. A student argues, "Model B has a lower in-sample MSPE, so it will predict outcomes more accurately for new observations." Do you agree with this conclusion? Explain.

2. This question is based on our review of statistics. A company advertises that its wireless earbuds last an average of 7 hours on a single charge. You independently test a random sample of $n = 64$ pairs of this model under the same conditions. Let b_i denote battery life in hours. Your sample has a mean of $\bar{b} = 6.6$ hours and a variance of $\text{var}\,(b) = 4$ hours squared. The sample distribution of the sample mean is therefore $\bar{b} \sim \mathcal{N}(\mu, 1/16)$, where μ is the true average battery life.

   (a) Form a 95% confidence interval for the population mean battery life. Interpret this interval in words.

   (b) You suspect that the average battery life is less than the advertised 7 hours. Test the null hypothesis that $H_0 : \mu = 7$ at the 5% significance level.

3. The following questions will focus on a dataset from Spotify. A random sample of songs were drawn from 6 genres: 'EDM', 'latin', 'r&b', 'pop', 'rap', and 'rock'. Spotify produced a few variables we will use for the analysis: (1) `track_popularity` measures track popularty on a scale from 0 to 100 and (2) `danceability` measures how suitable a track is for dancing on a scale from 0 to 1.

   I ran a regression of track popularity on the song's danceability measure.

```
OLS estimation, Dep. Var.: track_popularity
Observations: 32,833
             Estimate Std. Error t value  Pr(>|t|)
(Intercept)   35.1757   0.632777 55.5894 < 2.2e-16 ***
danceability  11.1497   0.945354 11.7942 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

   i. Interpret the coefficient on a song's `danceability` measure

   ii. Form a 95% confidence interval for the coefficient.

4. Continuing with the Spotify dataset, we want to see if different genre's have different popularities. We regress the track's popularity on indicators for each genre (excluding one)

```
OLS estimation, Dep. Var.: track_popularity
Observations: 32,833
               Estimate Std. Error   t value  Pr(>|t|)
(Intercept)   47.744870   0.339019 140.83227 < 2.2e-16 ***
genre::edm   -12.911344   0.451279 -28.61054 < 2.2e-16 ***
genre::latin  -0.718294   0.490232  -1.46521   0.14287
genre::r&b    -6.521339   0.488259 -13.35632 < 2.2e-16 ***
genre::rap    -4.529416   0.457639  -9.89736 < 2.2e-16 ***
genre::rock   -6.016532   0.489295 -12.29632 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

   i. What is the omitted group in this regression?
   ii. Interpret the coefficient on rock in words.
   iii. Which genre is more popular, r&b or rap?
   iv. How would you modify this regression to test whether this difference is statistically significant?